

Load-Aware Hierarchical Information-centric Routing for Large-scale LEO Satellite Networks

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Abstract—The emerging large-scale low earth orbit (LEO) constellation is expected to provide global Internet services. However, large-scale satellite networks meet the challenges of topology dynamics and continuous traffic variation. This paper proposes LoHi, a Load-aware Hierarchical Information-centric (LoHi) Routing Protocol based on constellation Partitioning and logic path identifier (PID), to address the above challenges. Specifically, LoHi divides a constellation into satellite groups and only keeps track of the inter-satellite link (ISL) state within each group instead of the entire constellation, stabilizing global routing. Moreover, LoHi adopts the PID to denote the logic connectivity of adjacent satellite groups, which corresponds to multiple physical ISLs. Accordingly, the hierarchical multipath routing is calculated based on the inner-group and inter-group connectivity information. Furthermore, LoHi adjusts forwarding decisions according to the real-time traffic load on the ISL and group-level inter-group paths to reduce packet drop and queuing delay in a hierarchical pattern. Packet-level experiment results show that LoHi achieves a higher packet delivery ratio than the state-of-the-art mechanisms (up to 132.12%).

Index Terms—Satellite networks, routing protocol, logic path identifier, load balance.

I. INTRODUCTION

A. Background & Motivation

Large-scale low earth orbit (LEO) satellite networks are becoming a vital communication infrastructure that provides low-latency and broadband Internet services [1]. The LEO constellations deployed by SpaceX and OneWeb are promising to support many delay-sensitive Internet applications, e.g., live video streaming, remote sensing, and real-time communication. This goal requires reliable content delivery within the LEO satellite constellation via the inter-satellite links (ISLs) [2]. Therefore, it is crucial to develop an appropriate intra-domain routing architecture and design a reliable routing mechanism for content delivery in large-scale constellations.

The challenges of achieving reliable content delivery (i.e., high packet delivery ratio) in large-scale LEO constellations are two-fold: 1) dynamic topology caused by intermittent ISLs; 2) unbalanced traffic distribution on the constellation. On the

one hand, the regular and occasional ISL failure renders the satellite network topology frequently change [3]. On the other hand, the Geo-distributed population yields unbalanced traffic from the global perspective, which leads to an unbalanced load for large-scale satellite networks [4]–[6]. Overcoming these challenges requires an appropriate routing architecture.

The classic IP-based (i.e., host-centric) routing architecture adopts an end-to-end communication model, which is vulnerable to topology changes. Moreover, the end-to-end communication model is also inefficient in distributing popular contents (e.g., popular live video) due to redundant delivery [7]. Hence, the classic IP-based routing architecture may not be able to accommodate the large-scale satellite networks. In contrast, information-centric network (ICN) routing architecture focuses on contents (instead of hosts). The named contents can be cached and re-used for subsequent requests, enabling reliable content sharing. However, the information-centric routing architecture cannot accommodate the intermittent ISLs, since it follows stateful forwarding. This will lead to a poor performance for the end-to-end traffic pattern.

The discussions above indicate that directly adopting existing terrestrial routing architectures in large-scale LEO satellite networks is not feasible. Therefore, it is necessary to investigate the topology feature and customize a proper routing architecture for this brand-new network circumstance. To this end, this paper proposes a load-aware hierarchical routing architecture for large-scale LEO satellite networks by integrating the advantages of IP-based and ICN-based routing paradigms.

B. Related Works

We review the literature closely related to our study in this article. They include the routing mechanisms that deal with satellite networks' topology dynamics and load balance.

1) *Routing mechanisms on dealing with topology dynamics*: Several studies integrate the topology control strategy (i.e., virtual topology method) with the link-state routing in satellite networks to deal with the occasional ISL failure (e.g., [8], [9]). For example, Pan *et al.* in [8] integrate the link-state detection with the regular routing table updates to reduce the onboard satellite memory. Moreover, some studies have explored topology management strategy (i.e., constellation partitioning) to

This work was supported by National Natural Science Foundation of China under Grants 62202021, 62225201, and 62271019, in part by the State Key Laboratory of Software Development Environment under Grant SKLSDE-2022ZX-18.

stabilize the global topology, yielding a hierarchical routing pattern [10], [11]. For example, Zhang *et al.* in [10] leverage the topology feature to divide the constellation into multiple areas and restrict the ISL state change within the area to tackle the strong topology dynamics.

2) *Routing mechanisms for on-board load balance:* The above routing mechanisms could get relatively good results when the network traffic is light. But their performance will decrease as traffic increase. Accordingly, several studies examine the problem of load balancing [4]–[6], [9], [12]. For example, Taleb *et al.* in [4] consider the congestion status among neighboring satellites when it selects the forwarding ISL. Song *et al.* in [5] propose a traffic-light satellite routing strategy to reduce packet drop and distribute traffic burdens fairly based on the ISL congestion status. However, the above routing mechanisms still suffer frequent routing updates, decreasing the content delivery performance under large-scale constellations, especially when ISL failure occurs.

Different from the literature above, our study in this paper will customize a proper load-aware routing architecture for large-scale satellite networks by integrating the benefits of IP-based and ICN-based routing paradigm.

C. Main Results and Key Contributions

This paper presents a *Load-aware Hierarchical Information-centric (LoHi) Routing Protocol* for achieving reliable content delivery in large-scale LEO satellite networks. Our main results and key contributions are as follows:

- *Propose a hierarchical information-centric routing architecture LoHi, enabling harness advantages of host-centric and information-centric routing:* LoHi reduces the topology dynamics by partitioning the constellation into several satellite groups, and each LoHi satellite only keeps track of the ISL state within each satellite group. Moreover, LoHi identifies the logic path of adjacent satellite groups, and adopts a hierarchical routing scheme (based on in-packet path identifiers).
- *Design a load-aware hierarchical content retrieval pattern:* LoHi introduces a load-aware hierarchical content retrieval scheme, releasing the advantages of host-centric and information-centric routing. Specifically, LoHi adopts a pull-based inter-group content retrieval by sending GET packets (information-centric) that record the inter-group PID of traversed satellite groups. Moreover, LoHi employs an push-based (host-centric) inner-group routing based on the in-packet PID. Particularly, LoHi can adjust forwarding decision according to the real-time traffic load on inter-group logic path and ISLs (i.e., select a properly next-group or next-hop satellite).
- *Packet-level performance evaluation:* We evaluate the performance of our proposed LoHi under the OneWeb constellation. Results show that LoHi outperforms the state-of-the-art routing mechanisms up to 132.12% in terms of packet delivery ratio.

The remainder of the paper is organized as follows. Section II elaborates the key idea of LoHi. Section III presents

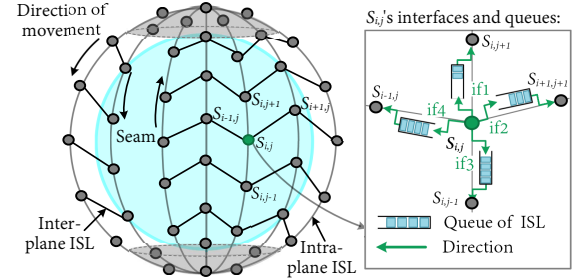


Fig. 1. Illustration of polar constellation and ISL's queue.

the design of LoHi in details. Section IV evaluates the performance of LoHi. Section V concludes this paper.

II. LOHI ROUTING ARCHITECTURE

A. Satellite Network Model

We consider a polar orbit constellation with N orbital planes and M satellites on each plane. The satellites are orbiting at the same altitude. Let $\{S_{i,1}, S_{i,2}, \dots, S_{i,M}\}$ denote the satellites on the i -th plane. The satellites in the same orbit plane are spaced uniformly, i.e., the angular distance between two adjacent satellites is $\omega = 2\pi/M$. Overall, satellite networks exhibit a deterministic neighbour relationship. Moreover, each satellite has at most four neighbors, which are connected via ISL (e.g., laser ISLs). As shown in Fig. 1, the satellite $S_{i,j}$ has four neighbors $\{S_{i-1,j}, S_{i,j-1}, S_{i,j+1}, S_{i+1,j}\}$. Furthermore, we suppose that there is a queue for each interface to buffer pending forwarding packets. As shown in Fig. 1, the satellite $S_{1,5}$ has four queues.

B. Key Idea of LoHi

LoHi aims to develop a routing mechanism that considers routing stabilization and load balance for satellite networks.

1) *Topology Management:* In terms of topology management, our proposed LoHi achieves the global topology stability via the following two approaches:

- **Constellation Partition:** We divide the constellation into multiple satellite groups, and the ISL state will be only disseminated within the corresponding group. Specifically, each group consists of p orbital planes and s satellites on each orbital plane. The satellite groups have deterministic neighbor relations. The neighbor groups are mutually connected by p intra-orbit ISLs or s inter-orbit ISLs. In practice, the group scale (p, s) could be flexible, which might be determined by the network operator.
- **Logic Connection:** We introduce the path identifier (PID) to represent the logic path between adjacent groups. These PIDs will be used in the inter-group content request and data delivery. Generally, they will not be advertised throughout the network and will be known only by the two adjacent groups. Moreover, a PID is associated with multiple physical ISLs, which is used to carry out multipath inter-group routing and improve the stability of the inter-group content forwarding.

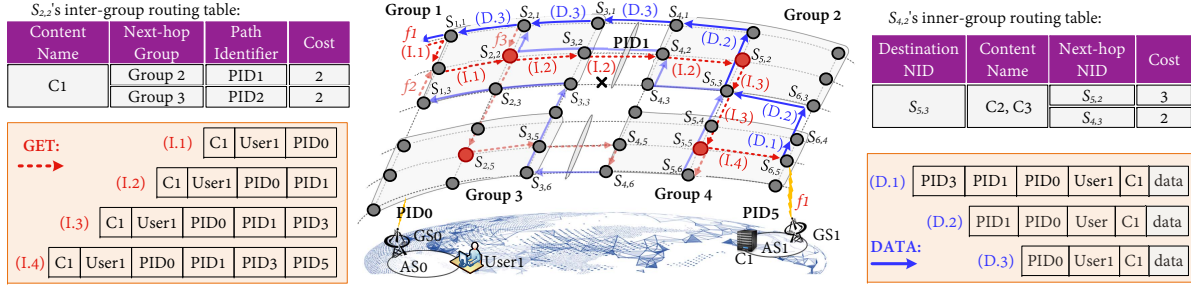


Fig. 2. Illustration of how LoHi routing mechanism works.

Fig. 2 plots a simple scenario with four satellite groups, each of which consists of nine satellites. Note that PID0 identifies the path between Group 1 and terrestrial AS0.

2) *Hierarchical Multipath Routing Calculation*: LoHi decouples inner-group routing and inter-group routing to construct a hierarchical multipath routing mechanism by leveraging the constellation partitioning and the PID.

- **Inner-group routing**: Each LoHi node detects the ISL state via the hello mechanism and synchronizes the inner-group *ISL State Advertisement* (LSA) within corresponding groups rather than the entire network. This yields less protocol overhead and improves the routing stability.
- **Inter-group routing**: The inter-group routing will construct a group-level topology and generate an inter-group routing table on each management satellite. Such an inter-group routing table will be used for guiding the group-level content retrieval in a pulling manner (like NDN).

3) *Hierarchical Load-aware Packet Forwarding Adjustment*: LoHi is information-centric and assigns each content a unique name with a hierarchical structure. According to the content name, LoHi adopts a hierarchical pull-based inter-group and push-based inner-group routing to fetch the desired content. Moreover, LoHi adjusts the forwarding decision in a hierarchical pattern with the following two aspects:

- **Inter-group packet forwarding**: Note that the management satellite will append the PID on the GET packet (i.e., request) to indicate the next-hop group towards the content provider. The inter-group traffic can be estimated in the management satellite by collecting content meta-data (e.g., content size) in each GET packet. As a result, the management satellite can adjust the forwarding decision, i.e., next-hop group. Thus, we could achieve load-aware multipath inter-group routing by properly sending GET packets along different next group. Accordingly, the DATA will be delivered back to the user following the reversed group-level path of forwarding the GET packet.
- **Inner-group packet forwarding**: In LoHi, the packets (i.e., GET/DATA) will be pushed to a proper destination within the group based on the inner-group routing and real-time ISL's queuing status. Particularly, the DATA packets will be forwarded within the group based on the in-packet PID and real-time ISL's queuing status.

C. An Illustrative Example

We show how LoHi works with the help of Fig. 2. Specifically, a User1 in the autonomous system AS₀ requests the content C₁ provided by a Provider in AS₁ via the satellite network. The content fetch under LoHi follows two stages.

GET Packet Forwarding: The red arrows in Fig. 2 represent the GET packet forwarding in the constellation. The User1 in AS₀ desires to obtain the content C₁, thus it issues a GET packet containing the name of C₁. The ground station (i.e., GS₀) in AS₀ will encode PID0 into the GET packet and forward the modified GET packet to the coverage satellite S_{1,1}. As illustrated by the red arrow (I.1) in Fig. 2, when S_{1,1} receives the GET packet, it will make a forwarding decision (to the management satellite S_{2,2} of Group 1 based on its **inner-group routing table and ISLs' load status**. Then, S_{2,2} will select a proper next group (i.e., Group 2) **according to the inter-group routing table (as shown in Fig. 2) and estimated inter-group traffic status**. Then, S_{2,2} appends PID1 to the end of the GET packet and forwards it to one of the egress border satellites (i.e., S_{3,2}), as illustrated by arrow (I.2). Similarly, as shown by arrows (I.3) and (I.4) in Fig. 2, the GET packet is forwarded to the satellite S_{6,5}, which covers GS₁ in AS₁. While the Provider in AS₁ receives the GET packet, it then delivers the DATA packet to User1 in the following procedure.

DATA Packet Forwarding: The blue arrows in Fig. 2 represent the DATA packet forwarding process based on the PID sequence recorded in the GET packet. Specifically, the Provider encodes the obtained PID sequence of the GET packet into the DATA packet. Then, the DATA packet will be forwarded to S_{6,5} via GS₁. As shown by the blue arrow (D.1) in Fig. 2, S_{6,5} forwards the DATA packet to the nearest border satellite S_{6,4} associated with PID3. If one of the ISLs with a heavy load in S_{5,3}, it adjusts its forwarding decision based on the **inner-group routing table** (as shown in Fig. 2) and **ISLs' load status**. Similarly, the satellites in Group 2 will forward the DATA packet based on the encoded PID1 and adjusted forwarding decision, as illustrated by arrow (D.2) in Fig. 2. In particular, if the inter-group ISL failure occurs (i.e., the black cross among border satellites S_{3,3} and S_{4,3}), S_{4,3} is capable of forwarding the DATA packet to Group 1 via an alternative inter-group ISL since each PID is associated with multiple physical ISLs. Finally, when the DATA packet reaches Group 1, the border satellite S_{1,1} will forward the DATA packet to

the GS_0 via the logic path PID0.

So far, we have introduced our key ideas of LoHi. Next, the LoHi routing algorithm is presented in detail.

III. LOHI ROUTING ALGORITHM

In this section, we first introduce how to build a hierarchical multipath routing. Then, we describe how LoHi adjusts forwarding decision based on real-time conditions.

A. Hierarchical Multipath Routing Calculation

We detailed elaborate the hierarchical multipath routing mechanism based on the constellation partitioning as follows.

1) *Detection of Physical Link Status*: To deal with links dynamics in the context of rapidly moving satellites, LoHi dynamically adopts the Hello mechanism to detect the link state (i.e., link failure and recovery).

- **Inner-group link state detection**: In our proposed LoHi routing mechanism, the satellite and ground-station (GS) will detect the connectivity of the ISL and ground-satellite link (GSL) status based on a hello protocol. Therefore, LoHi satellites can keep the ISL and GSL connectivity information up-to-date. When an inner-group link state change is detected, the LoHi satellite will initiate an *inner-group LSA* and announce the inner-group LSA with the group. For example, as shown in Fig. 3(a), $S_{1,2}$ and $S_{2,2}$ will create an *inner-group LSA* for the ISL failure (denoted by the black cross), and disseminate the LSA within the group (denoted by the dash green arrows).
- **Inter-group link state detection**: Similarly, the inter-group link changes will trigger the LoHi satellite (i.e., border satellite) to generate an *inter-group LSA*, which contains the adjacent group ID of the neighboring satellite. For example, as shown in Fig. 3(a), the LoHi satellite $S_{3,3}$ and $S_{4,3}$ will create an *inter-group LSA* for the inter-group ISL failure (denoted by the red cross), and disseminate the LSA with the corresponding groups.

2) *Detection of Logic Inter-group Path Status*: The management satellite will collect the connection relationship among the adjacent groups via the inter-group LSA (generated by border satellites) to maintain the logical connectivity of the inter-group. When all of the physical links between two groups are disrupted simultaneously, the logical relationship between two adjacent groups changes. Accordingly, the associate management satellites will create an *inter-group path LSA*. For instance, as shown in Fig. 3(b), when the ISLs among Group 1 and Group 2 are corrupted, border satellites $S_{3,1}$, $S_{3,2}$, $S_{3,3}$, $S_{4,1}$, $S_{4,2}$, and $S_{4,3}$ will generate an *inter-group LSA* with affiliated groups, respectively. Accordingly, when management satellites (i.e., $S_{2,2}$ and $S_{5,2}$) detect the logic inter-group path failure, they will initiate an *inter-group path LSA* and disseminate it to adjacent groups.

3) *Announcement of Physical ISL and Logic Inter-group Path Status*: Once an LSA (including inner-group LSA, inter-group LSA, or logic inter-group path LSA) is received, it will be announced within the corresponding groups. Specifically,

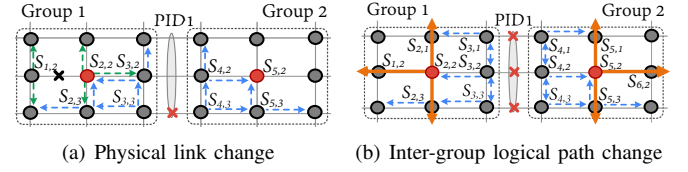


Fig. 3. Illustration of physical ISL and inter-group logical connection change.

• Inner-group and Inter-group LSA synchronization:

When the LoHi satellite receives an *inner-group LSA*, it will update its *inner-group LSDB* and disseminate the LSA to adjacent satellites within the group. Similarly, the *inter-group LSA* will be announced within the corresponding two groups. Particularly, each inner-group node will update the mapping of PID and the border satellite after receiving the *inter-group LSA*. For example, as shown in Fig. 3(b), the border satellite $S_{3,3}$ and $S_{4,3}$ will create an *inter-group LSA* for the inter-group ISL failure and disseminate the LSA with the corresponding groups (denoted by blue arrows). Such an LSA announce manner can efficiently propagate link state changes within the satellite group and yield a small protocol overhead, improving global routing stability.

- **Logic inter-group path LSA synchronization**: When the logic inter-group path change is detected, it will be advertised to the management satellite of each group, as illustrated by orange arrows in Fig. 3(b). This enables the management satellite to maintain up-to-date information on the logic inter-group path and calculate the inter-group routing table to guide packet forwarding.

Moreover, each satellite in a group maintains a mapping from the PID to the border satellites, which incurs little overhead.

4) *Multipath Routing Calculation*: LoHi node uses the collected adjacency information (i.e., inner-group LSA or logic inter-group path LSA) to construct a hierarchical multipath routing. Specifically, each inner-group node uses the collected inner-group LSA in the corresponding LSDB to build an inner-group routing table. Note that an extension of the shortest path first algorithm is employed to calculate the inner-group multipath routing. More specifically, the satellite first gets the number of links, which is the number of calculation iterations. Then, the satellite selects one of the links to calculate the cost to reach every inner-group satellite using Dijkstra's algorithm. This process is repeated for every available ISL. Note that the satellite also ranks the next hops for each satellite based on their respective cost. A similar group-level inter-group routing calculation is conducted.

Next, we present how the LoHi adjusts the forwarding direction based on the real-time traffic status.

B. Real-time Packet Forwarding Decision

Based on the above hierarchical multipath routing, LoHi is capable of adjusting the forwarding direction according to the real-time traffic load in a hierarchical pattern.

1) *Inner-group Forwarding Path Rating*: The inner-group forwarding path will consider the propagation and queuing

delays to find an appropriate path. In particular, the queuing delay can be calculated based on the output interface's real-time queue length, as shown in Fig. 1. To calculate the path rating, we will calculate the ISL cost of each interface in satellite $S_{i,j}$. Specifically, at time t , the instant ISL cost of interface k in $S_{i,j}$ is defined as

$$LC_k(t) = T_{pk}(t) + T_{qk}(t) + T_{Sk}(t), \quad (1)$$

where, $T_{pk}(t)$ denotes the one-way propagation delay among the adjacent satellite. $T_{qk}(t)$ denotes a predicted queuing delay in ISL k at time t , and it is computed as follows:

$$T_{qk}(t) = \frac{\sum_{k=1}^K L_k}{BW}, \quad (2)$$

where, $\sum_{k=1}^K L_k$ denotes the current queue size. BW is the bandwidth of an ISL. Moreover, T_{Sk} denotes routing detour for the packet delivery with an exponentially distributed.

Based on the above design, a satellite will rate the multiple outgoing paths for forwarding decision. Thus, the satellite will select the least-cost ISL to forward GET or DATA packet.

2) *Inter-group Forwarding Path Rating via estimating Logic Path Loading*: Recall that in our proposed LoHi, the group-level traffic loads can be estimated (i.e., traffic matrix G_a) in the management satellite since the GET will be sent to it for querying the next forwarding group. Accordingly, the predicted number of retrieved DATA packets N_g could be calculated. This motivates us to calculate the expected inter-group path cost at the group level. Let T_g denote the inter-group path cost for next-group forwarding, which is computed as follows:

$$T_g = \frac{N_g \times L_{avg}}{A_{links} \times BW}, \quad (3)$$

where, L_{avg} is the length of the packet. Moreover, A_{links} is the number of available ISLs among the adjacent satellite group.

3) *Hierarchical Load-balancing Forwarding Algorithm*: Algorithm 1 presents the major procedure of forwarding decisions in LoHi. It mainly consists of two key parts:

- **Inter-group path selection in management satellite:** Lines 1-6 present the progress of packet forwarding when the management satellite receives a GET packet. Specifically, the management satellite will select the least-loaded inter-group logic path to forward the GET packet and append the associated PID (i.e., based on the PID to determine the next group).
- **Inner-group ISL selection:** Lines 7-10 present how a LoHi satellite processes the received packets (GET and DATA). Specifically, the pre-calculated inner-group routing table will be used to provide a rough direction (i.e., Line 7). Further, the forwarding adjustment is carried out to help make the best decision (i.e., Lines 8-10).

IV. PERFORMANCE EVALUATIONS

To simulate the satellite network and conduct packet-level experiments, we develop a satellite network simulation platform based on OMNeT++. This simulation platform consists of three key modules, including the constellation module

Algorithm 1: Hierarchical Load-balancing Forwarding

Output: Interface $intf$

```

/* Process in Management Satellite */
1 if GET packet then
2   Lookup Inter-group content routing table
3   Select the next Group ID with minimal inter-group
   logic path cost  $Tg_a$  to the destination Group
4   Encode the PID associated with next Group ID
   into the GET packet
5   Send the Get packet to next group  $g$ 
6   Update traffic matrix  $G_a$  and logic path cost  $Tg_a$ 
   in the Inter-group content routing table
/* Process in Forwarding Satellite */
7 Lookup Inner-group routing table
8 Calculate available routes cost  $LC_k$  to the destination
9 Select the  $intf_k$  with minimal cost  $LC_k$ 
10 Send the packet to the output interface  $intf_k$ 

```

(to simulate the satellite mobility), the network module (to simulate the protocol stacks), and the traffic generator module (to simulate traffic patterns). Moreover, we implement LoHi and conduct extensive packet-level experiments to demonstrate the performance of LoHi and compare it with the state-of-the-art methods (i.e., NLSR [13], NDN-RFA [14], and LPIH [11]).

In our simulation, we consider the OneWeb constellation, which is a polar-orbit constellation with inclination 87.9°. It consists 720 satellites that fly along 18 orbital planes (i.e., 40 satellites on each orbit plane). Moreover, we suppose that each satellite has at most four ISLs (i.e., intra-plane ISLs and inter-plane ISLs). The payload of each DATA packet is 1 KB. For simulation convenience, we consider 10 Mbps for all links (i.e., ISL and GSL). Drop-Tail-based buffer of each link equals 500 packets. Furthermore, we simulate occasional ISL failures by randomly generating the failure events according to a Poisson process with a specific rate λ . Each failed ISL will recover after τ seconds.

A. Performance Under Stable Network Traffic Flows

We evaluate the packet delivery performance achieved by NLSR, NDN-RFA, LPIH, and LoHi under the OneWeb constellation. Specifically, the 6×10 grouping scheme of LPIH and LoHi is adopted, i.e., 12 groups with 60 satellites per group. We focus on the average packet delivery ratio (PDR) with constant request rates (i.e., stable network traffic flows) under an end-to-end traffic pattern. Now, we explore how the traffic load and ISL failure rate affect the content delivery performance of the LoHi routing mechanism.

Impact of Traffic Loads: Fig. 4 shows the results with different traffic loads¹ under point-to-point content delivery pattern. The horizontal axis represents traffic loads. From Fig. 4, we find that LoHi outperforms the state-of-the-art mechanisms, which implies that LoHi can accommodate the heavier

¹The traffic load is characterized by the number of requests per second in our experiments (i.e., 500 packets per second).

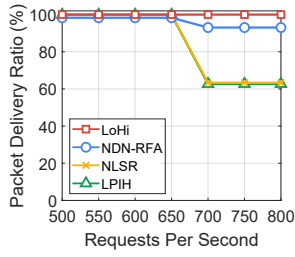
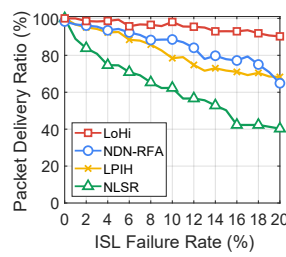
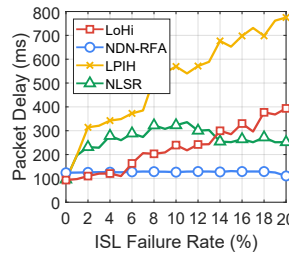


Fig. 4. Avg. PDR under different traffic load without ISL failure



(a) Avg. packet delivery ratio



(b) Avg. packet delay

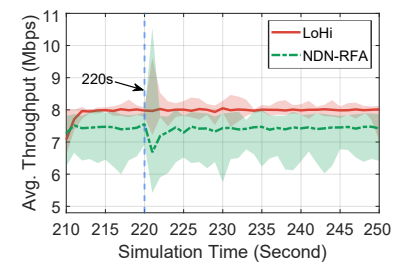


Fig. 6. Performance of LoHi and NDN-RFA under burst network traffic flows

traffic load. Specifically, LoHi achieves 7.49%, 59.56%, and 57.94% higher average PDR than NDN-RFA, NLSR, and LPIH, respectively. This is because the LoHi can adjust forwarding decisions based on the real-time traffic load on each ISL to avoid transient queues and packet drops.

Impact of ISL Failure Rates: We evaluate whether our proposed LoHi routing mechanism performs well in the large-scale satellite network with unreliable ISLs. Fig. 5 shows the results with different ISL failure rates (i.e., $\{0, 1, \dots, 20\}$). Particularly, we only consider the successful packet delivery in terms of packet delay.

Fig. 5(a) shows that as the ISL failure rate increases, the average PDR achieved by the above mechanisms decreases. Nevertheless, the performance of LoHi still outperforms the state-of-the-art mechanisms. More specifically, the average PDR of LoHi is higher than that of NDN-RFA, NLSR, and LPIH, up to 39.14%, 132.12%, and 32.22%, respectively. The reasons are two-fold. First, LoHi can mitigate the topology dynamics of satellite networks based on PID, improving packet forwarding reliability. Second, the LoHi can adjust packet forwarding decisions based on the real-time traffic load on ISLs to reduce the packet drop. As for packet delay, Fig. 5(b) shows that the packet delay of the above mechanisms increases as the ISL failure rate. This is because the traffic shift occurs when ISL fails, resulting in packet queues.

B. Performance Under Burst Network Traffic

Now, we compare LoHi and NDN-RFA (i.e., NDN-based load balance method) to evaluate their performance in dealing with burst traffic. Fig. 6 shows the average throughput with shaded areas, indicating the range of fluctuations when one of the traffic flows generates a burst of content requests. Particularly, the traffic burst occurs at time 220s.

By comparing the red solid and green dashed lines in Fig. 6, we can see that LoHi achieves a higher average throughput than that of NDN-RFA (up to 19.2%). Moreover, from Fig. 6, we find that LoHi has a slight variation in throughput performance, implying LoHi achieves a stable throughput than NDN-RFA. This is because LoHi can adjust the forwarding decision based on the real-time ISLs' load.

V. CONCLUSION

In this paper, we presented the design of the Hierarchical Load Balanced Routing (LoHi) for large-scale LEO satellite

networks. We also evaluate the content delivery performance of LoHi in terms of point-to-point content delivery ratio and instantaneous throughput by extensive packet-level experiments. Our results indicate that the LoHi accommodates the heavier traffic load. Moreover, our method outperforms the state-of-the-art mechanisms in terms of instantaneous throughput under the bursty nature of network traffic. In the future, we will further improve LoHi by considering the special demands on content delivery delay or bandwidth.

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